Experiment 18

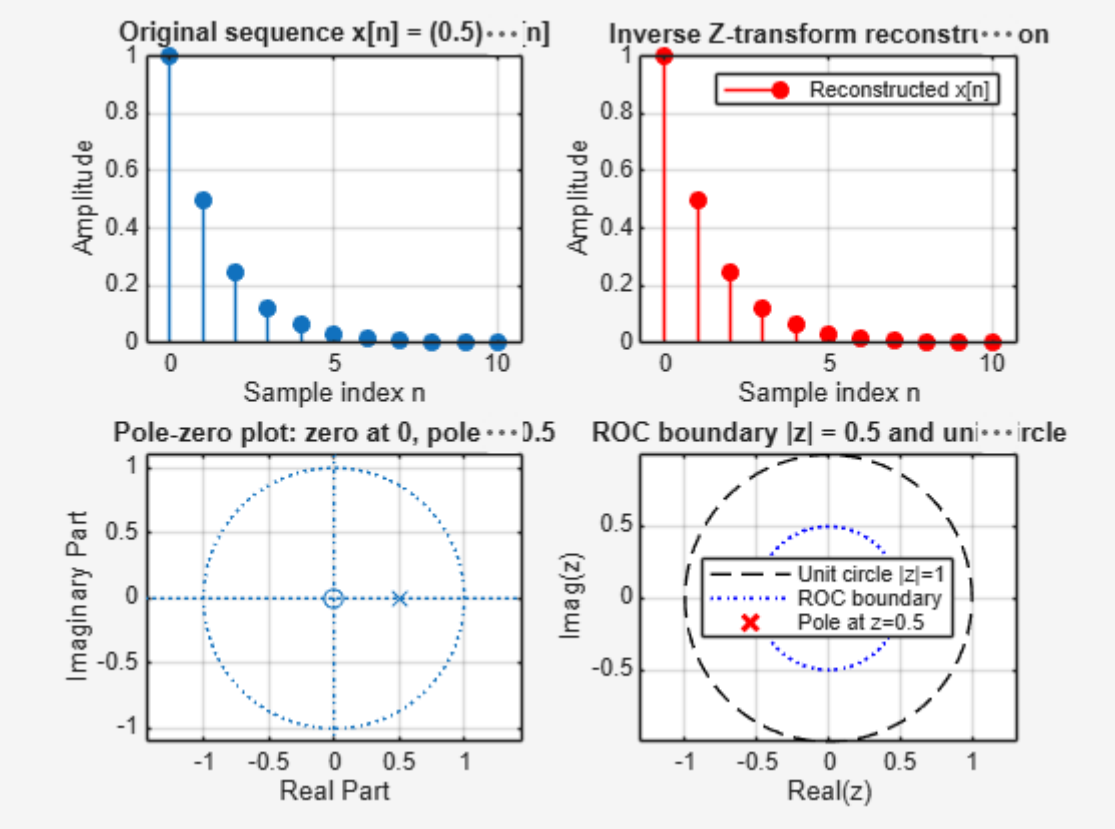
Z -Transform, ROC and Inverse Z - Transform

```
clc; clear; close all;
% x[n] = (0.5)^n u[n], a right-sided sequence
n = 0:10;
x = (0.5).^n;
% Analytical Z-transform: X(z) = z/(z - 0.5)
% ROC for the right-sided infinite sequence: |z| > 0.5
syms z k
Xz = z/(z - 0.5);
xInverse = iztrans(Xz, z, k);
disp('Z-transform X(z) = ');
pretty(Xz);
disp('Inverse Z-transform (symbolic) = ');
pretty(xInverse);
% Numerical inverse sequence for comparison over n = 0:10
xRec = (0.5).^n;
fprintf('Maximum reconstruction error = %.3e\n', max(abs(x-xRec)));
fprintf('Pole = 0.5; zero = 0; ROC for right-sided sequence: |z| > 0.5\n');
figure('Name','Experiment 18: Z-transform and inverse');
subplot(2,2,1);
stem(n,x,'filled'); grid on;
xlabel('Sample index n'); ylabel('Amplitude');
title('Original sequence x[n] = (0.5)^n u[n]');
subplot(2,2,2);
stem(n,xRec,'r','filled'); grid on;
xlabel('Sample index n'); ylabel('Amplitude');
title('Inverse Z-transform reconstruction');
legend('Reconstructed x[n]');
subplot(2,2,3);
zplane([1 0],[1 -0.5]); grid on;
title('Pole-zero plot: zero at 0, pole at 0.5');
subplot(2,2,4);
theta = linspace(0,2*pi,500);
plot(cos(theta),sin(theta),'k--','LineWidth',1); hold on;
plot(0.5*cos(theta),0.5*sin(theta),'b:','LineWidth',1.2);
plot(0.5,0,'rx','MarkerSize',10,'LineWidth',2);
axis equal; grid on;
xlabel('Real(z)'); ylabel('Imag(z)');
title('ROC boundary |z| = 0.5 and unit circle');
legend('Unit circle |z|=1','ROC boundary','Pole at z=0.5','Location','best');
```

CODE

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Z -Transform, ROC and Inverse Z - Transform



Output